

Proposition 1 (Global sensitivity of the RBF utility on the unit sphere $\mathcal{S}^{d-1} \subseteq \mathbb{R}^d$). *Let $\mathcal{V} \subset \mathcal{S}^{d-1}$ and consider the utility function $u_{rbf}(\cdot, \cdot)$*

$$u_{rbf}(e, v_i) = \exp(-\gamma \|e - v_i\|^2), \quad \forall v_i \in \mathcal{V}, \quad \gamma > 0,$$

where $e, v_i \in \mathcal{S}^{d-1}$ (the unit sphere). Then the global sensitivity of u_{rbf} is

$$\Delta u_{rbf} = 1 - \exp(-4\gamma).$$

Proof. By definition, the global sensitivity of u_{rbf} is

$$\Delta u_{rbf} = \max_{v_i \in \mathcal{V}} \sup_{e, e' \in \mathcal{S}^{d-1}} |u_{rbf}(e, v_i) - u_{rbf}(e', v_i)|. \quad (1)$$

Since u_{rbf} is continuous on the compact set $\mathcal{S}^{d-1}$, the supremum is attained, and

$$\Delta u_{rbf} = \max_{v_i \in \mathcal{V}} \left(\max_{e \in \mathcal{S}^{d-1}} u_{rbf}(e, v_i) - \min_{e \in \mathcal{S}^{d-1}} u_{rbf}(e, v_i) \right). \quad (2)$$

Expansion of the squared distance. For arbitrary $e, v_i \in \mathbb{R}^d$,

$$\|e - v_i\|^2 = \langle e - v_i, e - v_i \rangle = \|e\|^2 - 2\langle e, v_i \rangle + \|v_i\|^2.$$

Using the unit-sphere assumption. Since $e, v_i \in \mathcal{S}^{d-1}$, we have $\|e\|^2 = \|v_i\|^2 = 1$, hence

$$\|e - v_i\|^2 = 2 - 2\langle e, v_i \rangle = 2(1 - \langle e, v_i \rangle). \quad (3)$$

Utility Reformulation. Substituting (3) into the definition of u_{rbf} ,

$$u_{rbf}(e, v_i) = \exp(-2\gamma(1 - \langle e, v_i \rangle)). \quad (4)$$

Range of the inner product. By the Cauchy–Schwarz inequality,

$$|\langle e, v_i \rangle| \leq \|e\| \cdot \|v_i\| = 1,$$

so $\langle e, v_i \rangle \in [-1, 1]$. Moreover, both bounds are reached in $\mathcal{S}^{d-1}$:

$$\langle e, v_i \rangle = 1 \text{ for } e = v_i, \quad \langle e, v_i \rangle = -1 \text{ for } e = -v_i.$$

Min and Max of the Utility. Since $\exp(\cdot)$ is strictly increasing and the argument in (4) is a strictly increasing function of $\langle e, v_i \rangle$, the extremes of $u_{rbf}(\cdot, \cdot)$ taking both inputs from $\mathcal{S}^{d-1}$ are attained at the extremes of the inner product, i.e., antipodal points. Therefore,

$$\max_{e \in \mathcal{S}^{d-1}} u_{rbf}(e, v_i) = \exp(-2\gamma(1 - 1)) = \exp(0) = 1, \quad (5)$$

$$\min_{e \in \mathcal{S}^{d-1}} u_{rbf}(e, v_i) = \exp(-2\gamma(1 - (-1))) = \exp(-4\gamma). \quad (6)$$

Conclusion. Substituting (5) and (6) into (2), and noting that the resulting quantity does not depend on v_i ,

$$\Delta u_{rbf} = \max_{v_i \in \mathcal{V}} (1 - \exp(-4\gamma)) = 1 - \exp(-4\gamma).$$

□